

Provenance-Aware Extraction of Structured Records from Heterogeneous Field Reports

*Maria Okonkwo, Devraj Patel, and Lena Sørensen
Institute for Estuarine Computing, University of Brackish Bay;
Coastal AI Lab, Nordfjord Technical University*

Abstract

Field ecology increasingly depends on unstructured PDF reports whose layouts vary by agency and decade. We present a pipeline that normalizes such reports into a shared JSON schema while attaching a verifiable provenance pointer to every extracted value. On a corpus of 1,240 wetland surveys the method reached 94.8 percent field-level accuracy and reduced manual verification time by 71 percent. We argue that provenance is not a convenience but a correctness requirement for downstream scientific use.

Keywords

structured extraction; provenance; document AI; ecological monitoring; schema normalization